Ujjwal Raj

Quantitative Developer

Quantitative Developer building statistical arbitrage and commodity analytics applications with Python, PostgreSQL and microservices. Experienced in FIX/TT, Kafka, TimescaleDB, REST APIs and React/TypeScript, with a foundation in quantitative finance and AI.

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WORK EXPERIENCE

Quantitative Developer

GemsCap Global Analyst Pvt. Ltd., Pune

Aug 2026 – Present

- **ROLE:** Currently developing statistical arbitrage applications. Built the FCA (Forward Curve Analytics) and HCM (Historical Curve Matching) apps end-to-end, helping traders see a commodity's current term structure curve and find where a similar trend played out in the past, using 40+ years of historical settlement data.
- **KEY ACHIEVEMENTS:** Recently led development of the FCA and HCM apps and the expired-contracts notification system. The notification system is live on the production server and alerts traders about expiring contracts with 99.9% reliability. FCA and HCM run in production with no lag.
- **TECHNICAL EXPERTISE USED IN DAY TO DAY WORK:** Experienced in both monolithic and distributed architectures. Apps are built in Python and PostgreSQL as microservices, with React and TypeScript on the frontend. Backend technologies used daily:
  - *APIs and Messaging*: REST APIs, Kafka event streaming, and both synchronous and asynchronous service communication.
  - *Caching and Data*: Redis, SQLite and in-memory caching for fast reads, and writing PostgreSQL and TimescaleDB queries for time-series data.
  - *Deployment and Monitoring*: Docker containers on k3s, with Prometheus and Grafana for monitoring.

Quantitative Developer Intern

GemsCap Global Analyst Pvt. Ltd., Pune

Jan 2026 – Jul 2026

- **ROLE:** Gained a working knowledge of derivatives and commodity markets (CME). Helped build the market-data pipeline over the FIX protocol with TT connectivity into the database, and assisted senior developers in building the Portfolio Management backend that tracks each trader's VaR, CVaR and open positions.
- **KEY ACHIEVEMENTS:** VaR and CVaR calculations delivered with 100% accuracy. Developed the settlement PnL calculation.
- **TECHNICAL EXPERTISE USED IN DAY TO DAY WORK:** Learned how quantitative decisions made by quant researchers are designed and implemented in production code, and worked in an Agile framework using Jira, Confluence and Bitbucket.
  - *FIX Protocol and TT*: market-data connectivity from exchange feeds to the database.
  - *Backend*: caching, Docker, k3s and REST APIs.

ACHIEVEMENTS

TECHgium 8th Edition Finalist

Top 36 teams among 40,000+ students.

Financial Mastermind, 2nd Prize

Built a money-allocation model that maximizes portfolio return for a given market situation.

SKILLS

Backend Development | System Design | API Development | Python | C++ | SQL | Microservices | Distributed Systems | React | TypeScript | JavaScript | OOP | Multithreading and Concurrency | API Gateway | FastAPI | Flask | Django | REST APIs | Kafka | FIX Protocol | PostgreSQL | TimescaleDB | Redis | Docker | Kubernetes | k3s | AWS | CI/CD | Prometheus | Grafana | Agile | Jira | Bitbucket | Confluence | Statistical Arbitrage | Time-Series Analysis | Machine Learning | LLM Applications | AI Agents | LangChain | PyTorch | TensorFlow | Scikit-learn | Computer Vision

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Kancheepuram

B.Tech. Mechanical Engineering

2022 – 2026 • CGPA: 8.1/10

PERSONAL PROJECTS

Option-Chain and Margin Calculator

Python application on the Upstox API that automates option-chain analysis, contract selection, premium analysis and margin calculation. **GitHub**

MLOps Pipeline for Network Security

Phishing-detection pipeline with FastAPI, MongoDB Atlas, Docker, GitHub Actions CI/CD, deployed on AWS EC2 and S3. **GitHub**

Chest X-Ray Pneumonia Classifier

VGG16 and custom CNN ensemble reaching 94.7% accuracy, served through a Flask app. **GitHub**

CERTIFICATIONS

- Amazon ML Summer School 2025
- NPTEL: Machine Learning
- NPTEL: Deep Learning
- Udemy: Complete Data Science, Machine Learning, DL, NLP Bootcamp